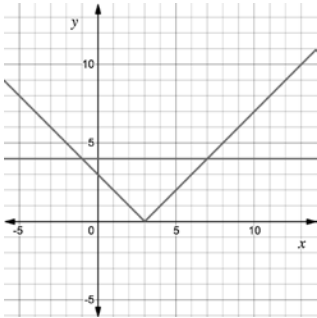


For questions 1-2, consider the absolute value equation $|x - 3| = 4$. The graphs of $y = |x - 3|$ and $y = 4$ are shown on the coordinate grid.



1. Use the graph to determine the solution to $|x - 3| = 4$.

$$x = -1 \text{ and } x = 7$$

2. Solve $|x - 3| = 4$ algebraically.

$$x - 3 = 4$$

$$+3 \quad +3$$

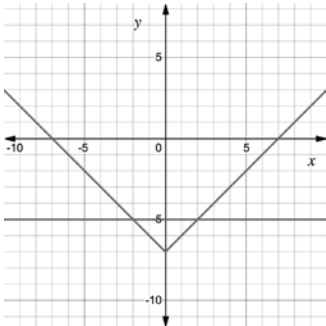
$$x = 7$$

$$x - 3 = -4$$

$$+3 \quad +3$$

$$x = -1$$

For questions 3-4, consider the absolute value equation $|x| - 7 = -5$. The graphs of $y = |x| - 7$ and $y = -5$ are shown on the coordinate grid.



3. Use the graph to determine the solution to $|x| - 7 = -5$.

$$x = -2 \text{ and } x = 2$$

4. Solve $|x| - 7 = -5$ algebraically.

$$|x| - 7 = -5$$

$$+7 \quad +7$$

$$|x| = 2$$

$$x = 2 \quad x = -2$$

For questions 5-10, solve each absolute value equation.

5. $|m| - 29 = -17$

$$+29 \quad +29$$

$$|m| = 12$$

$$m = 12 \quad m = -12$$

6. $|h - 16| = 21$

$$h - 16 = 21$$

$$h - 16 = -21$$

$$+16 \quad +16$$

$$+16 \quad +16$$

$$h = 37$$

$$h = -5$$

7. $|6x| = 21$

$$\frac{6x}{6} = \frac{21}{6} \quad \frac{6x}{6} = \frac{-21}{6}$$

$$x = 3.5 \quad x = -3.5$$

8. $11 = |x| + 8$

$$-8 \quad -8$$

$$|x| = 3$$

$$x = 3 \quad x = -3$$

9. $|k - 51| = 72$

$$k - 51 = 72 \quad k - 51 = -72$$

$$+51 \quad +51 \quad +51 \quad +51$$

$$k = 123 \quad k = -21$$

10. $20 = |8x|$

$$\frac{8x}{8} = \frac{20}{8} \quad \frac{8x}{8} = \frac{-20}{8}$$

$$x = 2.5 \quad x = -2.5$$